

PREHEATING PROSOCIAL BEHAVIOUR*

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We provide new evidence on motivations for voluntary contributions to public goods in a natural setting. Using natural language processing on users' Twitter posts, we measure revealed sentiment changes before and after making a donation to Wikipedia. We find strong evidence that sentiment improves in the hour and minutes directly before contributing, which we call 'preheating'. Results are robust to alternative fixed effects and approaches to inference and supported by a complementary online experiment with randomised mood inducement among Twitter users. Preheating suggests that affective states influence giving in addition to reward-seeking, utility-maximising behaviour that has been documented in other contexts.

Why do people contribute to public goods? Economists have long grappled with this question, developing and refining theories rooted in consequentialist social preferences (Bergstrom *et al.*, 1986; Andreoni, 1989, 1990), social norms and reciprocity (Fehr and Gächter, 2000; Bénabou and Tirole, 2006; Shang and Croson, 2009) and identity (Akerlof and Kranton, 2000; Bénabou and Tirole, 2011). Although these theories emphasise different motives for giving, they all conceptualise giving as reward-seeking behaviour. That is, contributing to public goods generates happiness or utility according to some underlying preference ordering. Yet an alternative explanation runs in the reverse direction—that one's *affective state* can spur giving—and this impetus has been the subject of much interest among psychologists (Adelman, 1972; Isen and Levin, 1972; Levin and Isen, 1975; Forgas *et al.*, 2008).

While not mutually exclusive, these differing conceptions of why people give are important to disentangle and analyse, both for illuminating the science of prosocial behaviour and because they have implications for policy and management. There are well-documented linkages between giving and happiness (Anik *et al.*, 2009; Lane, 2017), but ascertaining the direction of causality in real-world settings is difficult, and laboratory experiments often feature stylised giving tasks and/or artificially induced affective states, thus raising questions about external validity.

In this paper, we shed light on these questions by investigating changes in sentiment surrounding acts of charitable giving in the real world. Our work is novel in doing so in a natural setting using a revealed preference measure of individuals' well-being, and it also does so at scale for a large population. In particular, we use natural language processing to measure Twitter users' sentiment before and after donating to a public good (i.e., a monetary donation to Wikipedia). We analyse more than three million tweets and exploit within-user variation in revealed sentiment for more than 20,000 users who donated to the same public good over a six-month period.

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The data and codes for this paper are available on the Journal repository. They were checked for their ability to reproduce the results presented in the paper. The replication package for this paper is available at the following address: <https://doi.org/10.5281/zenodo.7915957>.

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Using an event study design based on the timing of donations, we find robust evidence of a *preheating* effect in which sentiment increases *prior* to donating, which is measurable up to an hour prior to giving. Our work is the first, to our knowledge, to identify such preheating in the moments immediately preceding a real-world prosocial action. We verify that our sentiment effects are indeed tied to the act of giving, as evidenced by a bootstrapped randomisation inference-style test with random event timing. We also demonstrate that our empirical framework identifies meaningful changes in sentiment by showing agreement among four independent sentiment lexicons and by our inability to find effects from randomly generated sentiment scores. The size and prevalence of the preheating effect suggest that transient moods and sentiment can shape giving decisions. The important role of affective states bears similarities with (some) nudges, which can affect economic behaviour without materially changing the choice set or consumption possibilities (Thaler and Sunstein, 2008).

We thus provide new evidence for why individuals engage in prosocial behaviour. Our approach has several attractive features. As a virtue of our research design, we observe both individuals' sentiment and voluntary charitable giving decisions in the wild. Our objective measures of sentiment are common across all individuals and we have a long panel, allowing us to identify effects using *within-person* comparisons. This strategy is a marked improvement over research designs that rely on comparisons of subjective measures across individuals or within-subject comparisons within a narrow experimental timeframe.¹ Furthermore, by studying a real-world social media setting, we avoid problems of verisimilitude that can arise when moods and/or prosocial behaviour are induced under artificial circumstances. We further probe this preheating effect by running an online experiment with the same focal population for our primary study, i.e., Twitter users who recently donated to a public good. This controlled experiment with induced moods complements our observational study by offering additional (albeit suggestive) evidence of a preheating effect and by helping assuage identification concerns regarding the timing, veracity of donations, and selection into tweeting about donating in our observational setting—issues that we discuss in greater depth below.

Distinguishing between motivations for public good provision is central to research in economics and psychology and crucial to efforts to stimulate charitable actions by governments and non-profit organisations. However, identifying underlying motivations for prosocial actions is difficult. Psychological studies tend to rely on subjective elicitations of motivations and affective states (Bischoff and Krauskopf, 2015; Lane, 2017). Meanwhile, economists have typically relied on changes in extrinsic motivations to *infer* the impact on intrinsic motivations through subsequent actions (Frey and Oberholzer-Gee, 1997; Chao, 2017). Seeking physiological evidence, neuroscientists and neuroeconomists have closely studied biochemical indicators, such as activity in certain regions of the brain, to deduce motives (Moll *et al.*, 2006; Fehr and Camerer, 2007; Harbaugh *et al.*, 2007; Mayr *et al.*, 2009).

Our work sits at the intersection of these different lines of research, providing distinct advantages in its own right. We study charitable giving in the wild, avoiding stylised giving environments and unrealistic or abstract scenarios that arise in many of the experimental interventions described above. Unlike the vast majority of work on this topic, our approach makes it possible to characterise sentiment and infer motivations at scale, something that would be infeasible for

¹ At the same time, we recognise several challenges incumbent in our approach, including sample selection and limitations to our donations data. Still, we see our approach as a promising and complementary way to shed new light on questions about motivations for engaging in prosocial behaviour.

studies of physiological indicators that require direct interactions with subjects and specialised equipment (e.g., fMRI machines).

Additionally, this research adds to a long literature in psychology and economics examining the relationship between happiness and prosocial behaviour. The literature has investigated both directions of causality (happiness increases charitability vs. prosocial behaviour increases happiness) in focused laboratory trials as well as large cross-country studies. Our preheating findings are in line with others that have found that happiness increases charitability (e.g., Aderman, 1972; Isen and Levin, 1972; Levin and Isen, 1975; Forgas *et al.*, 2008), while other studies have also investigated the reverse pathway through post-giving increases in sentiment (e.g., Dunn *et al.*, 2008; Aknin *et al.*, 2013; Geenen *et al.*, 2014; Bekkers *et al.*, 2019; Falk and Graeber, 2020). Unlike other work in this realm, we study moods and charitable actions that are naturally occurring rather than induced, and we do so by using within-subject comparisons for a considerably larger sample, providing new insights into this long-standing question. More broadly, our work relates to other research on how moods affect prosocial behaviours like co-operation and sanctioning (Drouvelis and Grosskopf, 2016) and reciprocity (Kirchsteiger *et al.*, 2006).

1. Observational Data and Empirical Setting

Our primary data source is built from information scraped from Twitter, the micro-blogging, social media platform. On Twitter, users have a profile and timeline where they view the ‘tweets’ or posts of other users.

We are primarily interested in users who have donated to charity. For this study, we focus on users who have donated to Wikipedia, which is an iconic example of a public good. As described on Wikipedia, Wikipedia is ‘a multilingual, web-based, free encyclopedia based on a model of openly editable and viewable content, a wiki. It is the largest and most popular general reference work on the World Wide Web, and one of the most popular websites by Alexa rank. It is owned and supported by the Wikimedia Foundation, a non-profit organisation that operates on money it receives from donors’.

To identify Wikipedia donors, we use Twitter’s public application programming interface (API) to search for users who tweeted the message: ‘*I just donated to Wikipedia. Support free knowledge! #iloveWikipedia*’, followed by a hyperlink to the donation website. When an individual completes an online donation, Wikipedia asks the donor to tweet this standardised message, and they make it easy to post the full message with a single click (see Figure A1 for screenshots of this process). We gather the list of users who tweeted this message (henceforth, the #iloveWikipedia tweet) at any point between 1 July 2018 and 31 December 2018, along with information about their social media account. We then scrape the entire history of tweets (excluding ‘retweets’, or unoriginal content) generated by those users before and for at least one month after the tweet containing #iloveWikipedia.² The public API returns a maximum of 3,200 tweets for each user. In total, our sample comprises tweet histories for 20,136 users who tweeted the verbatim #iloveWikipedia tweet described above.

Next, we perform sentiment analysis on the content of every tweet in our sample. Our sentiment analysis matches words in the tweet within a predetermined lexicon that translates the ‘sentiment’

² This scraping exercise was performed at the end of January 2019 and early February 2019, so we are limited to the universe of users’ tweets prior to that date.

Table 1. *Summary Statistics for Sentiment Variables and Users.*

	Mean	SD	Median	Min.	Max.	<i>N</i>
Standardised sentiment (AFINN lexicon)	−0.05	1.02	0	−30	26	686,337
Standardised sentiment (Bing lexicon)	−0.05	1.01	−0	−20	15	1,423,485
Standardised sentiment (NRC lexicon)	−0.04	1.01	0	−16	11	1,928,915
Standardised sentiment (VADER lexicon)	−0.04	1.00	−0	−3	2	3,095,357
No. of tweets	3,011.38	9,745.89	680	1	505,584	20,136
No. of followers	930.73	21,123.88	94	0	2,802,655	20,136
No. of likes (all tweets)	1,714.96	6,027.00	272	0	206,652	20,136

Notes: For the first four items, *N* represents the number of tweets scored under each sentiment lexicon. For the last three items, *N* represents the number of users included in the sample.

of tweets into a numeric variable, with positive numbers reflecting positive sentiment and negative numbers reflecting negative sentiment.³ The exception to this rule is VADER (Valence Aware Dictionary and Sentiment Reasoner), which embeds its own algorithm to determine the polarity (e.g., positive/negative sentiment) and intensity of the entire tweet text, providing a composite sentiment score. Expressed sentiment on social media, analysed via lexicon-based natural language processing, has been shown to correlate strongly with subjective well-being.⁴ Sentiment analysis using the AFINN, Bing, and NRC (National Research Council Canada) lexicons was performed using the ‘tidytext’ (version 0.3.2) package in R; analysis using the VADER lexicon was performed using the ‘vader’ (version 0.2.1) package in R. Lexicon dictionaries differ and some lexicons provide sentiment scores for words that others do not, which drives differences in sample sizes for each lexicon in our analysis (see Table 1). For all lexicons, we then standardise these scores to be mean zero with an SD of one for easy comparison and interpretation. Details on specific scoring and normalisation procedures are provided in Online Appendix A.1.

2. Empirical Strategy

We estimate changes in tweet sentiment immediately before and after individuals donate to the public good. To do this, we construct an event study, using the time that the user posted the #iloveWikipedia tweet as the event ($\Delta_{ihdm} = 0$). Let $S^L(\cdot)$ be a function that translates text into a standardised sentiment measure using lexicon *L*. We estimate our event study as follows,

$$S^L(\text{Tweet}_{ihdm}) = \alpha_{ih} + \sum_{b \in B} \gamma^b \mathbb{1}(\Delta_{ihdm} = b) + \sum_{a \in A} \gamma^a \mathbb{1}(\Delta_{ihdm} = a) + \lambda_d + \mu_m + \varepsilon_{ihdm}, \quad (1)$$

where Tweet_{ihdm} represents a tweet posted by user *i* in hour *h* of day *d* during month *m*. We include user by hour of day (α_{ih}), day of week (λ_d), and month (μ_m) fixed effects. We cluster our standard errors at the user level. As we will show, our results are robust to alternative specifications, fixed effects, and clustering.

³ Our approach is similar to previous studies that use sentiment analysis on social media content to measure changes in expressed sentiment as a result of changes in weather, air pollution and intensity of COVID-19 (e.g., Zheng *et al.*, 2019; Baylis, 2020; Wang *et al.*, 2020, 2022).

⁴ For example, Jaidka *et al.* (2020) find strong correspondence with county-level sentiment with standardised surveys of subjective well-being after removing common ‘stop words’. Similarly, Hao *et al.* (2014), Settanni and Marengo (2015) and Iacus *et al.* (2022) find significant correlations between sentiment of user-generated content on social media, processed through lexicon-based algorithms, and emotional well-being (via a standardised survey). Luhmann (2017) analyses a variety of studies to suggest that language patterns from social media posts correlate with measures of emotional experiences (but not life satisfaction).

We are primarily interested in the set of γ^b and γ^a coefficients, which captures the sentiment in specified time periods before ($b \in B$) and after ($a \in A$) tweeting. We explore ranges for B and A that include several hours before and after the #iloveWikipedia tweet, as well as 30- and 15-minute intervals before and after. Regardless of the timestep considered, we include a dummy variable that captures all periods after the event study period.⁵ We estimate (1) on a sample of tweets in a one-year (365 days) window before and after tweeting #iloveWikipedia, although the number of days after is censored by our sample collection procedure.⁶ We remove the standardised #iloveWikipedia tweet itself from the regression.

Given our rich set of fixed effects, γ coefficients are identified using within-individual variation in tweet sentiment at the same time of day, controlling for global day of week and month effects.⁷ Because we include a dummy variable for all ‘post’ time periods, all coefficient estimates are relative to the typical tweet that a user posts prior to the event study period within the sample bandwidth.

Our null hypothesis is that $\gamma^b = 0$ for all b and that $\gamma^a = 0$ for all a , i.e., that there is no relationship between sentiment and the act of giving. Evidence for $\gamma^a > 0$ would indicate that individuals experience positive sentiment after the act of charitable giving, consistent with a warm glow whereby individuals may derive hedonic mood benefits from the act of giving. Meanwhile, evidence for $\gamma^b > 0$ would indicate that positive sentiments *precede* giving, which we refer to as a preheating effect. Evidence of preheating would suggest the reverse direction of causality, i.e., an *affective* motivation in which positive sentiments cause charitable donations.

We acknowledge that our sample is highly selected. We have the universe of individuals who tweeted about donating to the Wikimedia Foundation in the last six months of 2018. There may be very good reasons why people who choose to tweet about giving to the Wikimedia Foundation differ from the general population of contributors to the same public good. In Table A3, we compare our sample to a set of Twitter users who tweeted ‘Wikipedia’ on 10 randomly chosen days in our sample period. We show that sentiment in our donor sample is comparable to other generic Twitter users, although our sample comprises less active and less popular Twitter users.

We stress that selection of this nature should not jeopardise inference. We rely on within-individual variation in sentiment to identify effects of giving, which nets out individual-specific heterogeneity that may be correlated with the propensity to give or the propensity to tweet about giving. Moreover, it is worth noting that empirical results based on lab or neurological experiments are also based on selected samples (e.g., college students) that may differ even more from the population of interest. Additionally, lab studies cannot credibly measure the anticipatory effects of *natural* charitable giving. In contrast, we observe sentiments and giving that occur in a natural setting.

⁵ For example, for four hours before/after, we set $b = (-4, -3], (-3, -2], (-2, -1], (-1, 0)$ and $a = (0, 1), [1, 2), [2, 3), [2, 3), [3, 4), \geq 4$.

⁶ User timelines were scraped at the end of January 2019 and early February 2019, so we only observe ~ 30 days of post-giving behaviour for donations made at the end of December.

⁷ To demonstrate where our variation is coming from, we plot the quantity of tweets and unique users within our event study time bins in Figure A2. Naturally, the quantity of tweets and users is largest in time periods immediately neighbouring the time of donation (i.e., we know a user is online and tweeting at $t = 0$, but the likelihood they are online decreases as we move away from the known time of the donation tweet). This analysis shows that we are able to identify our primary event study effects from hundreds of users who tweet frequently. We also see that VADER provides the largest sample of users and tweets, which helps explain why we see greater significance in our VADER results than with other lexicons.

3. Results

In this section, we report our primary empirical results. We then provide robustness checks and falsification tests to support our main results, followed by a discussion of potential threats to identification.

3.1. Primary Results

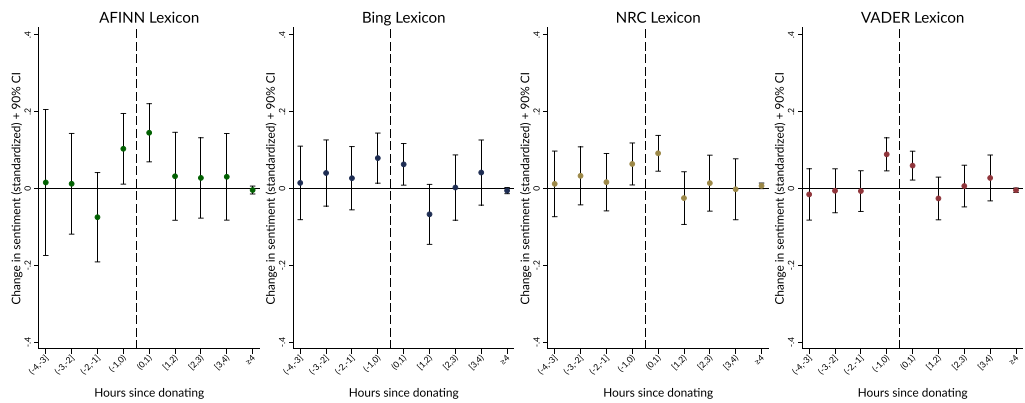
We present our primary results in Figure 1, plotting γ^a and γ^b coefficients and 90% confidence intervals from estimation of (1). Each panel presents coefficients from different time increments (one hour, 30 minutes and 15 minutes) before and after giving for our four lexicons. In panel (a) of Figure 1, the coefficient on the hour prior to donating is positive for each of our lexicons and remains positive in the hour after giving. The pre-giving effect ranges from 0.06 to 0.14 SD increases in sentiment across lexicons and each coefficient is significant ($p < 0.1$).⁸ This is our first evidence of a *preheating* effect. In the hour immediately following giving, coefficients are still positive (ranging from 0.06 to 0.14 SDs) and each coefficient is significant ($p < 0.1$). Outside this window, sentiment changes are not significantly different from zero and point estimates are small. Notably, the coefficients on sentiment of tweets beyond the event period (≥ 4 hours) are small and close to zero.

These results are sustained when estimated on finer time bins. In panel (b) of Figure 1, we focus on 30-minute intervals with an event window of two hours before/after donating. We observe similar results to the hourly intervals: preheating effects are positive for two 30-minute intervals prior to the #iloveWikipedia tweet, and remain statistically positive in the 30-minute interval following the donation tweet. Magnitudes are also comparable to the magnitudes of hourly effects in panel (a). Finally, in panel (c) of Figure 1, we repeat this event analysis on 15-minute time increments within the hour before and after the donation tweet. These coefficients are noisier, but we continue to find statistical evidence of a preheating effect during the hour prior to donating. Coefficients on some of these intervals exceed 0.2 SDs in magnitude. Post-giving effects, however, are generally smaller and dissipate within 15–30 minutes of the donation tweet. Beyond 30 minutes after donating, no coefficients are significantly different from zero.

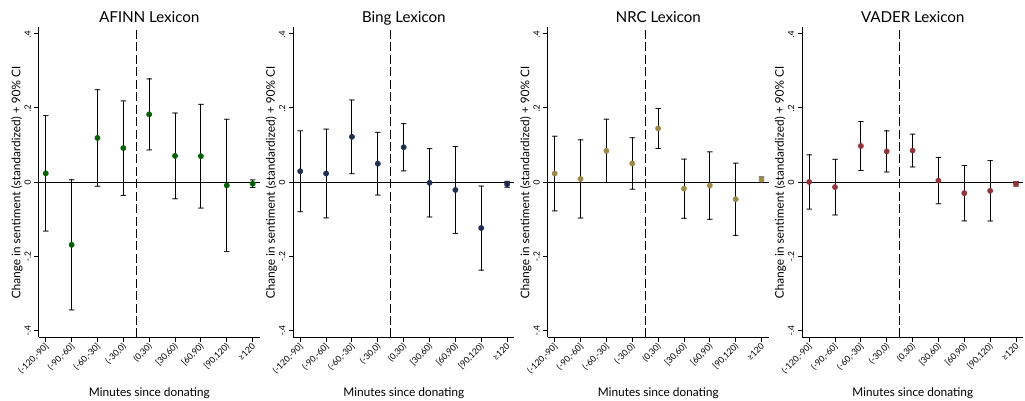
These results paint a clear picture: sentiment increases prior to tweeting #iloveWikipedia. We find that the sentiment of a user's tweets increases by roughly 0.1–0.2 SDs and persist through the donation. Increases in sentiment immediately before and after the donation tweet are comparable in magnitude. Beyond 30 minutes after the donation, however, we identify no changes in sentiment. Thus, we detect strong evidence of preheating, consistent with an affective motivation for giving. After that, we find more transient post-giving effects, which may be suggestive of warm glow derived from the donation, or alternatively, may be the artefact of sustained high sentiments from preheating. Regardless of the motivation, we find the quick dissipation of moods after giving to be indicative of short-lived changes in mood, suggesting that giving does not materially affect longer-run changes in well-being. Exploring these mood dynamics of giving would be an interesting line of future inquiry.

How big is a 0.1–0.2 SD change in sentiment? Our estimated effects are comparable in magnitude to the results in Baylis (2020) who examines sentiment changes in response to

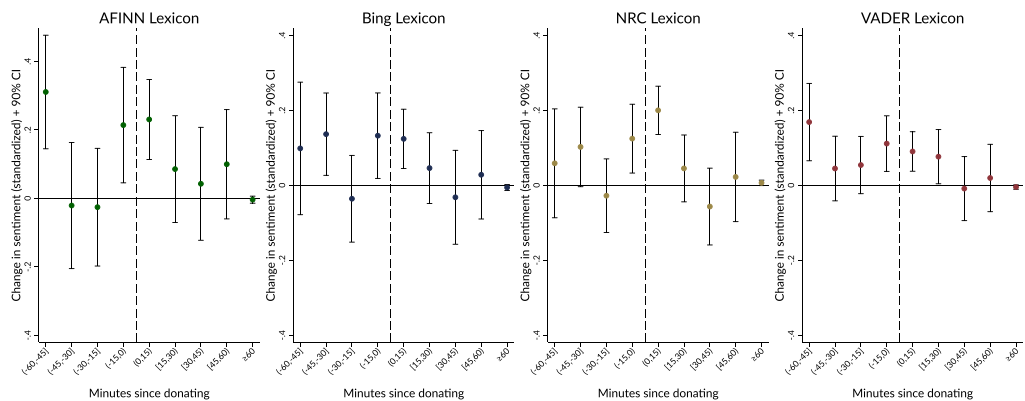
⁸ We present regression coefficients for hourly, 30-minute, and 15-minute intervals in the Online Appendix in Tables A4, A5, and A6. These tables include our preferred estimates shown in Figure 1 in columns (1), (5), (9) and (13). The remaining columns provide robustness checks for different fixed effects.



(a) Coefficient estimates for hour intervals before/after tweeting #iloveWikipedia



(b) Coefficient estimates for 30-minute intervals before/after tweeting #iloveWikipedia



(c) Coefficient estimates for 15-minute intervals before/after tweeting #iloveWikipedia

Fig. 1. Change in Sentiment for Hours and Minutes Before/after Tweeting #iloveWikipedia.
 Notes: Coefficients are estimates from equation (1), which includes user by hour, day of week, and month fixed effects. Standard errors are clustered at the user level.

temperature deviations. In that paper, exchanging one day with moderate temperatures (20–25°C) for an extremely hot day (i.e., maximum temperatures greater than 40°C) results in a 0.1–0.2 SD reduction in sentiment. In other words, our preheating effect is comparable in magnitude to reducing the maximum temperature by 15°C on a very hot day.⁹

3.2. Robustness Checks

To explore the robustness of our primary effects, we implement a randomisation inference-style procedure by randomising the timing of the donation event for each user and re-estimating our primary coefficients within a series of 500 bootstrap iterations (described in detail in Online Appendix A.2). We then compare our primary estimated coefficients to the empirical distribution of the bootstrapped coefficients, ensuring that our primary changes in sentiment are capturing a ‘behavioural’ response to the donation rather than picking up idiosyncratic variation in the data unrelated to the act of giving. Furthermore, this process provides verification that inference on our primary coefficients is accurate.

In Figure 2, we summarise our results for 30-minute intervals.¹⁰ Each column uses a different lexicon and each row presents a different 30-minute increment corresponding to our event study timing. There are several takeaways from this analysis. First, each empirical distribution of bootstrapped estimates based on random event timing is centred around zero for all time increments and all lexicons. The 90% confidence interval includes zero in all cases. This result suggests that idiosyncratic variation in our data does not explain our positive and significant treatment effects. Second, the significance of our estimates varies by timing in a way that is consistent with our primary analyses. For very early time periods (the 90- and 120-minute bins prior) and later time periods (after 30-minutes post), our primary coefficient estimates fall within the empirical 90% confidence intervals from the bootstrapped estimates. These coefficients are statistically similar to zero using estimated standard errors as well. For the 30- and 60-minute bins prior to donating and the 30-minute bin directly after donating, our primary coefficient estimates exceed all of the bootstrapped estimates. We view this result as strong evidence that our primary preheating and post-giving effects are driven by fundamental changes in sentiment associated with giving.

Additionally, we provide an extensive set of robustness checks in the Online Appendix. Our main results are robust to alternative sets of fixed effects that capture different heterogeneity over dates and weeks (Tables A4, A5 and A6) and to a local polynomial regression that exploits cross-sectional variation around the event time (Figure A3). We run another falsification test using false lexicons, with results in Figure A8 (more detailed description of this falsification test is provided in Online Appendix A.3). Tweets scored with these fake sentiment lexicons do not appear to respond to the timing of the #iloveWikipedia tweet, with the exception of a small, significantly negative effect for the hour prior to the event for the random lexicon. We view this as an artefact of noise in our data and it is consistent with the proportion of type I errors that we would expect to see in classical hypothesis testing.

⁹ In monetary terms, and perhaps more speculatively, Baylis (2020) calculates sentiment changes for tweets in response to changes in wages and receipt of parking tickets. Those results suggest a per-SD change in sentiment of \$196.77 and \$78.60, respectively.

¹⁰ For brevity, we present the same figures for our hourly and 15-minute intervals in the Online Appendix (Figures A6 and A7). The results tell the same story as the 30-minute results.

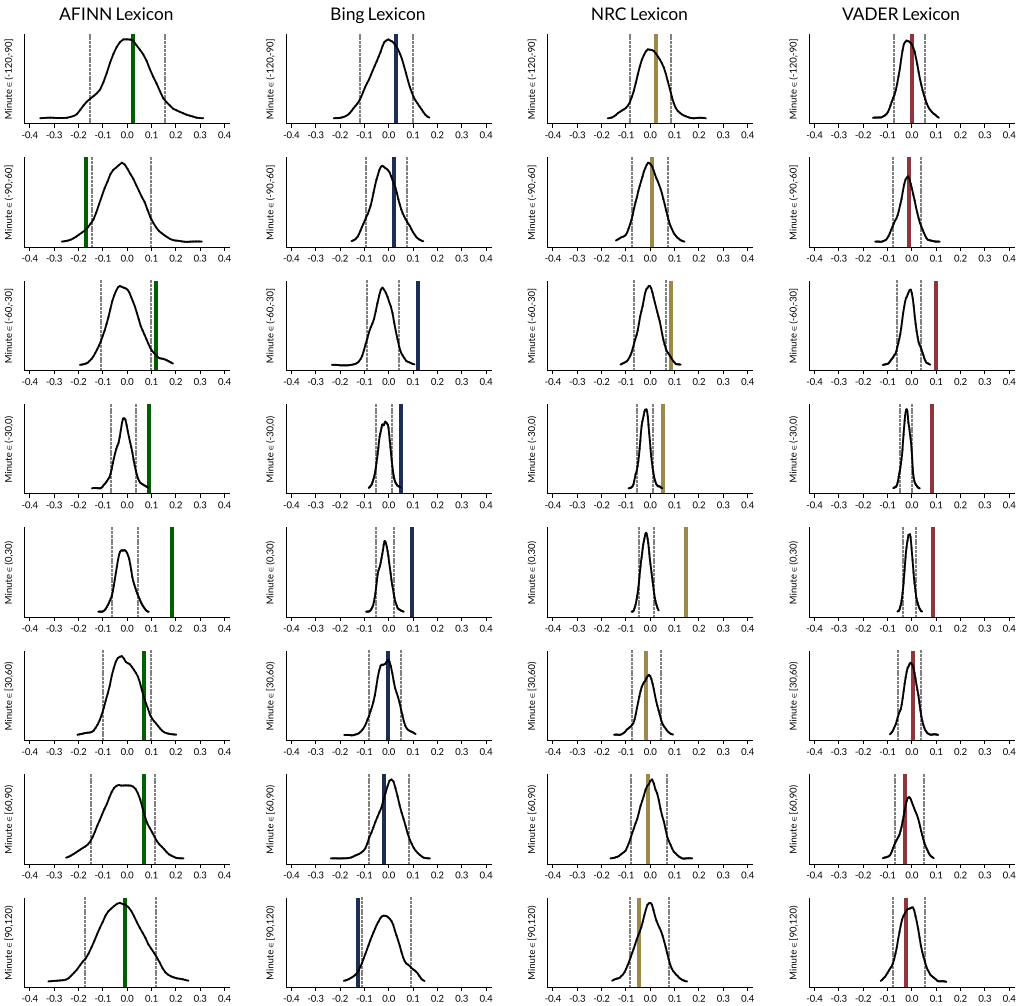


Fig. 2. *Randomisation Inference-Style Test for 30-Minute Intervals Since False Donation Time.*
Notes: In each figure, k-density distributions of parameter estimates are presented for 500 bootstrap repetitions (black curve). Dashed vertical lines represent the 5th and 95th percentiles (i.e., the empirical 90% confidence intervals) for the bootstrapped coefficient distribution. Bold vertical lines represent original parameter estimate. Each plot represents a distinct time period within the event study framework.

3.3. Interpretations and Caveats

Our results suggest strong preheating effects, i.e., that positive moods beget greater generosity. A core assumption of our analysis is that the #iloveWikipedia tweet corresponds with a donation to Wikipedia. However, without observing the actual donation itself, we face several potential threats to identification.

For one, it may be the case that a user posts the verbatim #iloveWikipedia tweet, but does not actually donate. While we cannot rule out such behaviour outright, we think it is unlikely to be prevalent enough to drive the results we are finding. First, such dishonest representation

takes some effort, especially to reproduce the verbatim text from Wikipedia's template on which we focus in our analysis. Second, it is unclear what the motive would be for (widespread) lying of this sort, particularly because our results suggest limited sentiment benefits after the #iloveWikipedia tweet is posted. Third, unless we make additional assumptions about how such dishonesty is correlated to pre and post sentiments, this type of dishonesty would tend to introduce measurement error and noise, which would attenuate estimates rather than create a spurious preheating effect.

Another challenge for identification is timing. There may be a lag between when the user actually donates and when they subsequently post the #iloveWikipedia tweet. Then, sentiment scores could potentially be driven by tweets occurring between the actual donation and the #iloveWikipedia tweet, in which case we are not identifying preheating effects for donative activity. This would require individuals to donate, navigate to Twitter to tweet about something other than the donation, and then navigate back to the donation browser to follow the prompts to share the #iloveWikipedia tweet. While some users may take this path, we strongly doubt that this is the overriding pattern of behaviour in our sample that is driving results. Furthermore, we believe that long time lags would, if anything, introduce noise into our estimates, as this would allow more time for the individual to experience different emotional triggers in between the actual donation and the announcement of the donation.¹¹

In case this potential time lag could actually introduce bias rather than simple noise, we run several robustness checks. We replicate our event study results for hourly and 30-minute time increments after removing tweets posted within the 5-minute interval prior to the #iloveWikipedia tweet, which we believe to be a rather long time lag for attention and activity in the social media world. Results are shown in the Online Appendix Figure A4; point estimates are similar to our primary results with larger confidence intervals. We further extend this robustness check for the 'hour before donation' coefficient. We run ten additional regressions, dropping successively larger intervals ranging from 3 minutes to 30 minutes prior to the #iloveWikipedia tweet, and report the results in the Online Appendix Figure A5. We find relatively stable, positive coefficient estimates for preheating. We see that the preheating effect remains, although significance varies across lexicons as we reduce the number of tweets that we are able to leverage for the analysis. Notably, all coefficients remain significant for the VADER lexicon, which is the lexicon that allows us to score the largest number of tweets.

Endogeneity and selection may also threaten clean identification. After making a donation, there is an additional voluntary (and potentially endogenous) decision about whether to tweet about the donation. We infer from our preheating effects that sentiments influence donations. However, an alternative explanation could produce similar looking preheating effects: perhaps users with positive sentiments are more likely to announce their donations online. Then, our identifying variation would come from differences in tweeting behaviour rather than differences in their actual generosity. Both explanations would be observationally equivalent under our research design, but we unfortunately do not have direct data on donations that would allow us to distinguish between these two channels in our observational study.

¹¹ A similar concern is that users may *decide to* donate at an earlier time, but they make their actual donation and subsequent tweet with a lag. Then, tweets in the interim may exhibit elevated mood, but that elevated mood would be attributable to the warm glow of deciding to donate rather than a true preheating effect. As with the pathway described above, we also believe that this is unlikely to be the overriding pattern of behaviour in our sample that is driving results, but we acknowledge that we cannot rule out this possibility definitively.

In light of this key challenge, we note that our conclusions about the impact of mood on donations is consistent with prior work finding the positive moods enhance other prosocial behaviours, like co-operation and sanctioning (Drouvelis and Grosskopf, 2016), reciprocity (Kirchsteiger *et al.*, 2006) and helping (Aderman, 1972; Isen and Levin, 1972; Williams and Shiaw, 1999; Forgas *et al.*, 2008). Moreover, these two channels are not mutually exclusive. Thus, if the ‘mood-on-tweet’ effect is indeed prevalent, this would inflate, but not necessarily invalidate, our inferred ‘mood-on-donation’ effect. Given that our primary interest is in the qualitative finding of preheating effects, we are less concerned about overstating the magnitude of these effects. However, there remains a valid concern about whether the preheating effects are truly indicative of greater donation activity. An unobserved, positive ‘mood-on-tweet’ effect could be driving the preheating results, rather than actual changes in donation activity amongst users with positive moods.

To address this challenge more directly, we design and run an experiment to complement our observational study. In this controlled setting, evidence for a preheating effect or evidence against a positive ‘mood-on-tweet’ effect would help corroborate our interpretation of preheating.

4. Experiment

4.1. Design and Implementation

Given challenges of disentangling donations from tweeting behaviour in our empirical setting, we design an incentivised online experiment that allows us to distinguish these channels. To maintain close comparability with our primary empirical work, we recruit participants from our original Twitter population.¹² We identify all users who posted the #iloveWikipedia hashtag in the calendar year of 2021, with some spillover into neighbouring months in 2020 and 2022. We then created two Twitter accounts (‘@GTResSurvey’ and ‘@GTSurvey’) with university branding for our research study and invited users to participate in the experiment through a direct message on Twitter. Our final sample is limited by several factors: the size of the population that posted the hashtag during this time frame, the fact that many users block direct messages from accounts that they do not follow, and response rates.¹³ Thus, we further expand the sample to include any users who have tweeted ‘I just donated’ in the same time frame. We include the ‘I just donated’ population, which is a superset of the original #iloveWikipedia tweet, to expand our recruitment population without sacrificing recency.¹⁴

The experimental design is simple in concept and follows prior economic studies that use mood induction (Capra, 2004; Kirchsteiger *et al.*, 2006; Capra *et al.*, 2010; Drichoutis and Nayga,

¹² In this way, we take a ‘lab-in-field’ approach (Gneezy and Imas, 2017; Gangadharan *et al.*, 2021), which is attractive because it ‘combines elements of both lab and field experiments in using standardised, validated paradigms from the lab in targeting relevant populations in naturalistic settings’ (Gneezy and Imas, 2017, p. 439).

¹³ Overall, we sent approximately 40,000 direct messages to Twitter users through an automated R script and roughly 11,500 of those messages were received by users; the remaining users did not receive the message because they block direct messages from users that they do not follow. Our response rate from the successfully transmitted messages was approximately 1.5%. Low response rates in this range are understandable given the nature of online recruitment, which many users believed to be a potential scam.

¹⁴ For practical reasons, we could not indiscriminately disseminate the survey. Doing so would invite rent-seeking behaviour from opportunistic users or bots, who fill out many surveys to earn larger payments. Thus, we limited the eligible participant group according to the user characteristics and time frames described above. In fact, 25 respondents completed the survey, but were not in our original recruitment pool, either because they found the survey link organically or because they failed to supply a Twitter handle that matched our recruitment list. As a result, we present results with and without these respondents.

2013; Drouvelis and Grosskopf, 2016). After providing informed consent, subjects completed an online Qualtrics form with four sets of tasks: (1) a video task to randomly induce a positive or neutral mood, (2) mood elicitation questions, (3) a charitable dictator game with real stakes, and (4) survey questions.

In the charitable dictator game, subjects were given an experimental endowment of \$50 to divide between themselves and a named charity. For users who tweeted the #iloveWikipedia hashtag, the named charity was the Wikimedia Foundation. For those who tweeted the generic 'I just donated' string, we first asked subjects to choose a preferred charity from a list that included Oxfam, Red Cross, Save the Children, the World Wildlife Fund, the Wikimedia Foundation and Doctors without Borders. Their choice would then become the named charity for the charitable dictator game task. This procedure follows prior work by Crumpler and Grossman (2008), Gangadharan *et al.* (2018) and Gandullia *et al.* (2020), among others.

Subjects were informed that 10% of surveys would be drawn for actual payment to the participant and charity. At the end of the survey, we also asked subjects to tweet about their experience using a template modelled after the #iloveWikipedia tweet: '*I just completed a research survey, and my participation may help provide financial support for charity. #supportresearch #supportcharity*'. This prompt is used to gauge potential selection into tweeting about prosocial behaviour.

We provide more detailed information about recruitment, the study sample, and experimental procedures in Online Appendix B.1. Full screenshots from the primary survey tasks are available in Online Appendix B.3.

4.2. Results

Figure 3 summarises results from the online experiment on Twitter, with summary statistics and regression tables available in Online Appendix B.2. As described previously, our final sample is limited by low contact rates, as many users block direct messages from unfamiliar accounts, and low response rates, which we speculate is because many users are (understandably) wary of clicking a survey link from an unfamiliar source.¹⁵

With this caveat in mind, we see higher average self-reported moods and feelings in the positive treatment (Figure 3, panels (a) and (b)), as well as higher average donation amounts (\$41.71 vs. \$38.89) (panel (c)) and higher likelihood of donating the full experimental budget of \$50 to charity (68% vs. 57%) (panel (d)). Participants in the positive treatment are, if anything, *less* likely to tweet about their participation than their counterparts in the negative treatment (20% vs. 27%) (panel (d)). However, while the differences in moods and feelings are significant ($p < 0.01$ for all specifications), the differences in donations are not ($p > 0.10$ across specifications). In our preferred specification, which is estimated on a restricted sample of respondents that we directly solicited on Twitter and controls for recruitment group, respondents in the positive mood inducement donated \$2.66 more than respondents in the neutral mood inducement (p -value = 0.26).¹⁶ Positive mood inducements also made respondents 10.4% more likely to donate all \$50 (p -value = 0.20). Unfortunately, given the limited sample size, it is difficult to ascertain

¹⁵ Our recruitment script was limited by Twitter's API to a maximum of 1,000 messages per day and one of our recruitment accounts was blocked permanently for being marked as spam.

¹⁶ This specification is preferred because it only analyses responses from users that we directly solicited on Twitter (thus excluding the 25 respondents who arrived at our survey by other means). It also controls for recruitment group, although results are essentially the same (and, if anything, a little stronger in magnitude) when we do not control for recruitment group.

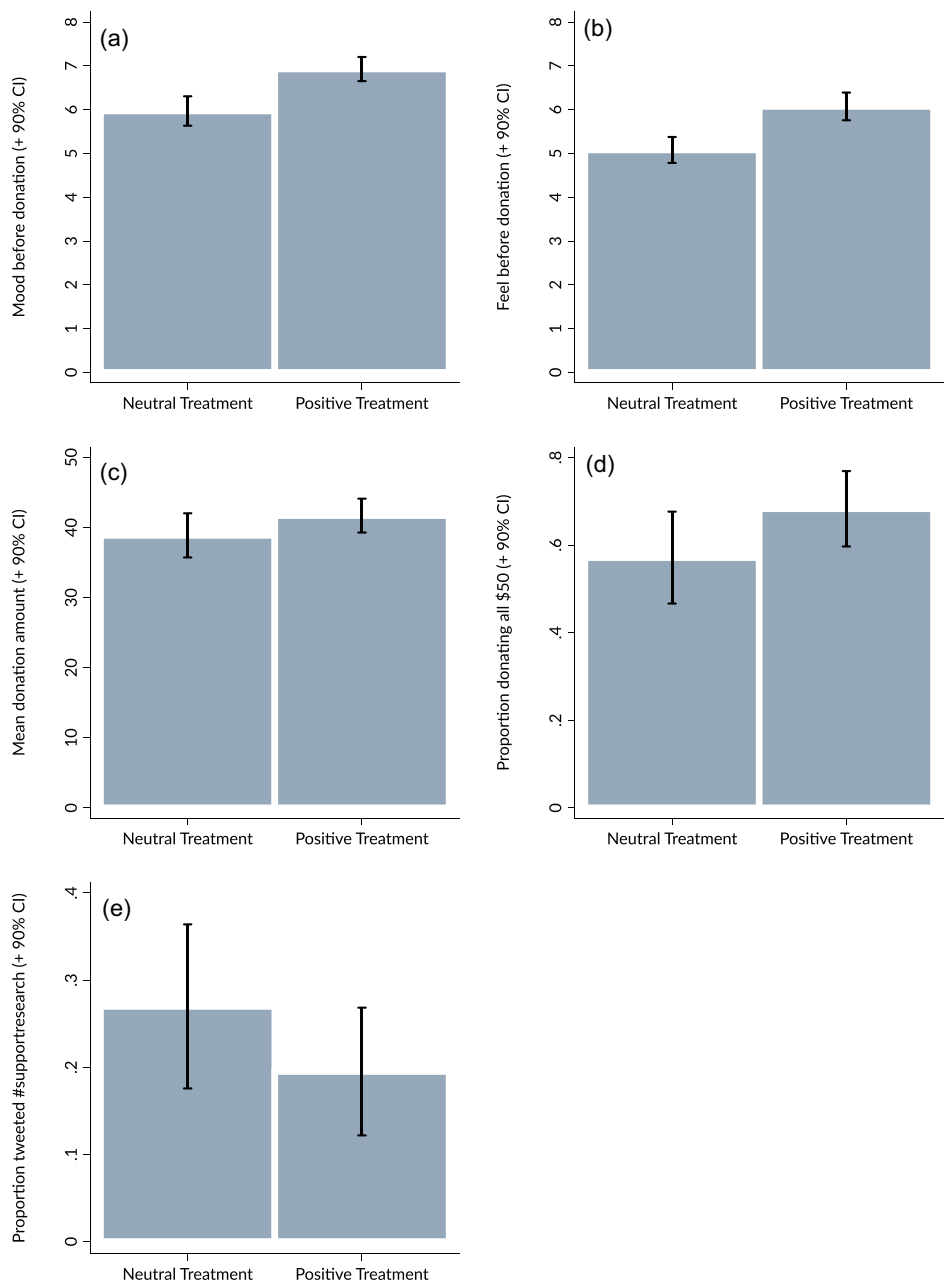


Fig. 3. Survey Results on Mood Inducement, Donation Amounts, and Tweet Behaviour.
Notes: Panel (a): Self-reported mood before donation (9-point scale from bad mood to good mood); Panel (b): Self-reported feeling before donation (9-point scale from depressed to elated); Panel (c): Mean donation amount; Panel (d): Proportion of respondents donating all \$50; Panel (e): Proportion of respondents tweeting #supportresearch prompt.

whether these are true null effects or whether the impact of positive sentiments is being masked by noise. It may be the case that the average differences in mood are not large enough to create statistically significant differences in the final outcome of interest (donations). Moreover, one might expect several important sources of noise in such an online experiment, e.g., if subjects are inattentive or distracted, if subjects doubt the likelihood of being paid, or if subjects undervalue stakes because they are playing with ‘house money’. These sources of noise will bias estimates toward zero and necessitate much larger treatment effects to be detectable at standard levels of significance.

Although the donation results are somewhat equivocal, the tweeting behaviour of experimental participants is interesting and informative for our observational study. We find a negative and statistically insignificant effect of mood on propensity to tweet about prosocial behaviour. Respondents receiving the positive mood inducement were 7 percentage points (or roughly 26%) *less* likely to tweet about supporting charity (p -value = 0.33). This effect lends support to our preheating interpretation from the observational study. There, a primary concern was that happier users might be more likely to post the #iloveWikipedia tweet, thus creating selection bias that could inflate our preheating effects. We find no evidence of such selection in the experiment. Instead, we find a statistically null effect, and the point estimate is negative; if anything, a negative correlation between mood and tweeting #iloveWikipedia would bias us *against* finding preheating effects in our primary observational analysis.

Our experimental results align with Kirchsteiger *et al.* (2006) and Drouvelis and Grosskopf (2016), which are two economic studies that also examine prosocial behaviours after a mood induction. Kirchsteiger *et al.* (2006) focus on reciprocity in a gift exchange game, finding greater generosity among participants with induced positive moods. Drouvelis and Grosskopf (2016) study co-operation and sanctioning behaviour in a public good game, reporting higher contributions from happy subjects and harsher punishments from angry subjects. Although these various behaviours may be related to charity, our work provides the most direct insights on real-world charity based on our game form and our study population. Moreover, our experimental findings are used as auxiliary evidence that dovetails with our results from observation of charitable giving ‘in the wild’, making our work unique from prior efforts in our realm. Our findings suggest that the preheating effect is consequential. We speculate that our online experiment would find similar impacts of sentiments if the aforementioned sources of noise could be diminished, and we pose this as an interesting avenue for future study.

5. Discussion and Conclusions

In this paper, we provide new evidence on an age-old question: what motivates prosocial behaviour? We use large-scale observational data from Twitter to shed light on the role of affective motivations in charitable giving. We find strong evidence of a *preheating* effect, i.e., a pronounced increase in sentiment immediately *before* the act of giving. This effect survives a battery of robustness checks—including different specifications, constellations of fixed effects, and methods for inference—and threats to clean identification are assuaged by further analyses and a complementary experimental study.

Overall, we find affect to be an important driver of prosocial behaviour, with potential ramifications for policy and management. For example, charities may raise more funds if they can solicit donors when they are ‘hot’, and there may thus be great value for detecting such sentiments in the wild and in real time. They may even benefit from directly inducing positive sentiments among

potential donors, in an effort to increase the likelihood and magnitude of donations. Indeed, psychology and marketing research has shown that consumer perceptions and decision-making are affected by moods, including those induced by the retail environment, and positive moods are associated with favourable product and store evaluations, purchase intentions and spending decisions (Babin and Darden, 1995; Sherman *et al.*, 1997; Spies *et al.*, 1997; Bakamitsos and Siomkos, 2004; Furnham and Milner, 2013). Our results point to parallel impacts in the realm of public good provision. An interesting avenue for future work is to explore how these affective motivations interact with other incentives for giving, such as matches, rebates and social information.

Along these lines, our findings have practical and conceptual implications for the classic ‘crowding out’ effect, whereby government provision of public goods may diminish private contributions. If people give to charity for the sake of augmenting a public good, then their contributions will be crowded out by government grants; however, such crowding out is mitigated if a warm glow is a primary motivation for giving (Andreoni, 1990; List, 2011). In a similar fashion, crowding out will be limited if donors are driven by affective motivations—another reason that real-world giving may deviate from the stark predictions of full crowding out.

To our knowledge, our work is the first to document preheating effects immediately preceding giving in a natural environment. While many researchers have investigated these phenomena using experimental interventions in the lab, our work is unique in identifying the role of sentiments and prosocial motivations in the wild. This distinction is a crucial one. As a virtue of our research design, we observe *revealed* sentiments and their relationship to *natural* prosocial actions in a real-world environment. Using unique observational data paired with newly developed natural language processing tools, we provide new insights on the important role of sentiments in shaping charitable behaviour.

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Additional Supporting Information may be found in the online version of this article:

Online Appendix Replication Package

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